

AKUL TYAGI

Delhi, India | akultyagi.cs@gmail.com | (+91) 7503898231 | [Portfolio](#): akul-tyagi.github.io
[LinkedIn](#): linkedin.com/in/akul-tyagi | [Github](#): github.com/Akul-Tyagi | Codeforces: 1540+ (Specialist-Top 7%)

PROFESSIONAL SUMMARY

Final-year B.Tech CSE student and Founder of Naivety - an offline-first Android ecosystem serving 1,000+ active users across 80+ countries with a verified 99.9% crash-free rate. Fine-tuned a production 7B LLM (QLoRA/4-bit quantization) powering 125,000+ impressions; 9,500+ engagements; reaching 43,000+ LinkedIn members. Active open-source contributor to Wikimedia Commons Android (2 PRs merged). One of the first globally to rebuild native C++ libraries for Android 15 kernel compliance. [Codeforces](#) Specialist in Java (1540+). Seeking SDE internship or full-time roles in Android, full-stack, AI or SWE.

TECHNICAL SKILLS

- Mobile Engineering: **Java, Kotlin, Android SDK, Jetpack Compose, Kotlin Coroutines, Room, Retrofit, Dagger-Hilt, MVVM**
- AI & Machine Learning: **Python, PyTorch, Hugging Face, LLM Fine-Tuning (QLoRA/LoRA), 4-bit Quantization, RAG.**
- Backend & Databases: **Node.js, FastAPI, Express.js, MongoDB, PostgreSQL, Redis, REST APIs, Distributed Systems.**
- Frontend Development: **React.js, Next.js (SSR/ISR), TypeScript, JavaScript, Three.js, TailwindCSS, Zustand.**
- DevOps & Cloud: **Docker, Kubernetes, AWS (EC2, S3), Firebase, CI/CD, Git, Linux/UNIX, Automated Testing.**
- Core Competencies: **Data Structures & Algorithms (Java), System Design, Object-Oriented Design (LLD), OS, Agile.**

EDUCATION

- Galgotias College of Engineering and Technology – B.Tech in Computer Science and Engineering (2023-2027)
- Academic Highlights: CBSE X - **93.2%** (2021) | CBSE XII - **82.8%** (2023) | JEE (Mains) - **89.1st** Percentile (2023)

WORK EXPERIENCE

Naivety (Incubated Startup–GCET Accelerator) | Founder & Lead Android Engineer | Nov 2025 - Present

- Agile built, shipped and iterated on an offline-first Android e-reader (EPUB/PDF) using Jetpack Compose, MVVM & Room Database - scaled to 1,000+ active users, 80+ countries, verified 99.9% crash-free rate from 30,000+ sessions ([Play Store](#))
- Forked & rebuilt custom C/C++ native module of an open-source PDF rendering engine([AndroidPdfViewer](#)-[PdfiumAndroid](#)) to enforce 16KB page-size alignment for Android 15 kernel compliance, reducing application memory footprint by 30%.
- Selected as 1 of 10 ventures from 300+ applicants for the GCET Startup Accelerator; received structured 1-on-1 mentorship on product strategy and scaling ([Naivety Video](#)).

Wikimedia Foundation | Open Source Contributor – Android | March 2026 – Present

- Merged 2 PRs ([#6752](#), [#6746](#)) in Wikimedia Commons Android app (millions of users); debugging critical upload pipeline bugs - infinite offline dialog loop in UploadMediaDetailFragment and a NestedScrollView scroll interception failure.
- Investigated & documented root causes for 5+ issues including blocking network call (CsrftokenClient.getTokenBlocking) causing ANR errors in Android Vitals; 3 additional PRs ([#6808](#), [#6765](#), [#6772](#)) in active maintainer review.
- Designed a comprehensive phased modularization strategy - migrating legacy RxJava/MVP architecture to Jetpack Compose, MVVM, and Kotlin Coroutines across feature:auth & feature:contributions modules (350-hr scope, [T420912](#)).

PROJECTS

Unagi – Full Stack AI Content Generation Platform | [Github](#) | [Live Link](#) | [Demo Video](#)

- Fine-tuned Falcon3-7B-Instruct on a custom-curated dataset of 1,200+ professional data points using QLoRA adapters and 4-bit bitsandbytes quantization; reduced inference latency by 35% on NVIDIA T4 GPUs.
- Deployed a FastAPI inference server with asyncio concurrency and ngrok tunneling; built a secure Next.js frontend with Supabase OAuth and a Three.js 3D interface, driving 125,000+ impressions reaching 43,000+ [LinkedIn](#) members.

CureAPT - Full-Stack Algorithmic Wellness & Data Engine | [Github](#) | [Live Link](#)

- Engineered a real-time ML algorithm processing 30+ diverse health inputs into a wellness index with O(1) constant-time lookups; built a reservoir sampling engine generating personalized feedback from 100+ actionable insights.
- Optimized MongoDB throughput for scalable sub-second data pipelines for edge deployment via global Mongoose connection pooling, preventing connection exhaustion during high-concurrency bursts.

CERTIFICATIONS

- **NPTEL - Deep Learning (IIT Ropar): Elite + Silver Medal** [Certificate](#) (75%) in 12-week course covering neural networks, optimisation (SGD/Adam), regularisation and autoencoders, PCA/SVD, CNNs, RNNs/LSTMs.